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Stability Issues in Sulfide-Based All-Solid-State Batteries: From Material Properties to Electrode Interfaces

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Stability Issues in Sulfide-Based All-Solid-State Batteries: From Material Properties to Electrode Interfaces

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Abstract:

All-solid-state batteries (ASSBs) are regarded as a pivotal technology for next-generation energy storage, given the high energy density and intrinsic safety. Among the various solid-state electrolytes (SEs), sulfide SEs are particularly attractive owing to their high ionic conductivity, good deformability, and favorable electrode contact. Nevertheless, their practical implementation remains constrained by pronounced stability issues arising from intrinsic material properties and complex electrode/electrolyte interfacial reactions. This review systematically summarizes recent progress in sulfide-based ASSBs, with emphasis on electrolyte instability and interfacial degradation mechanisms. The distinctive features of sulfide SEs are first outlined by comparing them with other classes of SEs. Subsequently, the origins of sulfide SE instability under air exposure, solvent contact, thermal stress, electrochemical operation, and mechanical loading are analyzed. Particular emphasis is placed on failure processes at electrode interfaces, including chemical and electrochemical reactions, chemo-mechanical coupling, and their impact on structural evolution and performance decay. Furthermore, recent interfacial engineering strategies, including electrode material optimization, electrolyte modification, interfacial coatings, and artificial solid electrolyte interphase formation, are discussed in detail. Finally, key challenges and future directions for achieving stable, high-energy-density sulfide-based ASSBs are proposed to guide materials design and practical implementation.

1 Introduction

As portable electronic devices, electric vehicles, and large-scale energy storage systems rapidly advance, lithium-ion batteries (LIBs) face more stringent requirements in terms of energy density, safety, and service life ^[1,2]. Traditional LIBs rely on flammable organic electrolytes, inevitably posing safety hazards (such as thermal runaway, electrolyte leakage, as well as combustion under high energy density conditions), and have gradually approached their theoretical performance limits ^[3,4]. Given this, all-solid-state batteries (ASSBs), which replace liquid electrolytes and commercial separators with solid-state electrolytes (SEs), are widely recognized as the central direction for next-generation energy-storage systems with high safety and high energy density ^[5,6]. ASSBs not only hold the promise of fundamentally eliminating the risk of electrolyte combustion but also enable compatibility with high-capacity electrode materials such as lithium metal anode (LMA), thereby significantly enhancing both energy density and safety.

As key components of ASSBs, the ionic conductivity, chemical stability, and compatibility with electrode materials of SEs directly determine the overall cell performance ^[7,8]. Currently reported SEs are primarily classified into oxides, polymers, sulfides, and halides. Oxide SEs typically exhibit high chemical and electrochemical stability, but their ionic conductivity at room temperature (RT) is relatively limited. Furthermore, oxide SEs suffer from brittleness and poor interfacial contact, necessitating high-temperature sintering or high-pressure assembly during fabrication ^[9,10]. Polymer electrolytes have good flexibility and interfacial wettability, but the ionic conductivity at RT and oxidation resistance are hard to meet the demands of high-energy-density batteries ^[11,12]. Halide SEs exhibit high ionic conductivity and exceptional chemical and electrochemical stability with high-voltage oxide cathode active materials (CAMs), but the poor compatibility with LMA, sensitivity to humidity, and high raw material and manufacturing costs limit practical applications ^[13,14]. In contrast, sulfide SEs are considered as one of the most promising SEs for practical application due to the high ionic conductivity at RT ($10^{-3}\sim 10^{-2}$ mS cm⁻¹, approaching

or even exceeding that of liquid electrolytes), excellent mechanical properties, and favorable interface compatibility ^[15-17]. Over the past decade, significant progress has been achieved in the research of sulfide SEs. Representative examples include glass and glass-ceramic systems ($x\text{Li}_2\text{S}-y\text{P}_2\text{S}_5$, $x+y=100$), LGPS-type material with chemical formula of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS), as well as the widely focused argyrodite-type sulfide SE with chemical formula of $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl) ^[18,19]. Through structural design, element substitution, and defect control, sulfide SEs achieve outstanding ionic transport properties at RT, forming the material foundation for constructing high-energy-density ASSBs. However, as research has gradually shifted from the intrinsic properties of materials to the device-level integration, a series of stability issues exhibited by sulfide SEs have been the core bottleneck hindering their further development ^[20-23].

On the one hand, sulfide SEs face significant stability challenges at the material level due to their inherent structural properties ^[24,25]. Compared to other inorganic SEs, the P-S and M-S bonds ($\text{M} = \text{Ge}, \text{Sn}, \text{Si}, \text{etc.}$) in sulfide SEs have lower thermodynamic stability, which causes extreme sensitivity to moisture and oxygen in the air ^[26,27]. Most sulfide SEs undergo rapid hydrolysis when exposed to humid air, producing H_2S gas alongside structural degradation and ionic conductivity reduction. This not only raises safety risks during material preparation and battery assembly, but also requires stringent conditions for large-scale production. Furthermore, sulfide SEs may suffer phase transitions or compositional rearrangement under high temperatures or prolonged operation, compromising their long-term structural stability and electrochemical performance ^[28]. Although sulfide SEs are generally considered to possess a wide theoretical electrochemical stability window, the actual stability is often significantly influenced by the interfacial environment ^[29]. Under high-voltage conditions, sulfide SEs readily oxidize and decompose to form insulating products such as elemental S, SO_4^{3-} , PO_4^{3-} , etc. Under low-voltage conditions, sulfide SEs are reduced by LMA to produce interfacial decomposition phases such as Li_2S and Li_3P . Such decomposition reactions are thermodynamically inevitable, yet the reaction kinetics, product composition, and interfacial evolution behavior are highly dependent on specific

material systems and interfacial structures ^[30]. These factors are critical determinants of battery cycling stability and the evolution of interfacial impedance.

On the other hand, the interfacial stability between sulfide SEs and electrode materials in ASSB systems presents more complex and challenging issues ^[22,31-33]. Unlike traditional liquid electrolytes, solid-solid interfaces lack spontaneous wetting and dynamic reconstruction capabilities, leading to the intercoupling of interface contact states, chemical reaction, and mechanical-matching issues. When matched with high-voltage oxide cathodes (such as layered NCM or lithium-rich cathodes), sulfide SEs are prone to interfacial oxidation decomposition during charging ^[34,35]. This leads to a rapid increase in interfacial impedance, severely limiting the rate performance and cycle life of the battery. Although coating or composite structures can mitigate interfacial reactions, the composition and thickness of the interfacial layer, as well as the impact on ion and electron transport at the interface, still require a systematic understanding ^[36]. Furthermore, significant interfacial instability issues also exist at the interface between sulfide SEs and LMA ^[37-39]. Sulfide SEs typically undergo reductive decomposition in contact with LMA, forming interfacial layers with complex chemical compositions ^[40]. While some decomposition products exhibit limited ionic conductivity and can partially function similarly to a solid electrolyte interphase (SEI), the mechanical integrity and long-term stability of the layer remain difficult to guarantee. During cycling, volume changes and stress concentration caused by lithium deposition/stripping readily induce cracking or failure of the interfacial layer, which in turn triggers lithium dendrite growth and internal short circuits ^[41]. In addition, maintaining stable interface contact under low or no external pressure conditions remains one of the critical challenges for achieving high-energy-density ASSBs.

In summary, despite demonstrating significant advantages in ionic conductivity and processability, the inherent instability of sulfide SEs and complex electrode interfacial reactions have become the critical issues that restrict the further development of sulfide-based ASSBs. It is necessary to systematically review and thoroughly summarize the stability challenges of sulfide SEs from multiple perspectives, including material

chemistry, structural design, and interfacial engineering. Therefore, this review centers on the core topic of stability, systematically summarizing the key challenges faced by sulfide-based ASSBs at both the electrolyte intrinsic properties and electrode interfaces (**Figure 1**). Firstly, from the perspective of structural characteristics, the intrinsic stability issues of sulfide SEs under air, thermal, and electrochemical environments are analyzed, as well as the regulation strategies. Subsequently, the interfacial reaction mechanisms between sulfide SEs and cathode/anode materials, interfacial failure modes, and the interfacial engineering methods proposed over recent years are discussed in detail. Finally, focusing on the future development directions and critical scientific challenges of sulfide-based ASSBs, this review provides valuable insights for designing highly stable, high-energy-density ASSBs.

2 All Solid-State Batteries

ASSBs feature solid materials for the cathode, anode, and electrolyte, representing the most promising next-generation electrochemical energy storage technology following liquid LIBs. Since the proposal in the 1950s, breakthroughs in SEs research have led to widespread global attention from both scientific research and industrial sectors due to the outstanding safety, high energy density, and long cycle life of ASSBs. Unlike traditional liquid LIBs, ASSBs replace organic SEs and separators with SEs, significantly simplifying battery integration and processing costs (**Figure 2a**) ^[1]. Furthermore, the SE simultaneously performs dual functions of ion conduction and electrode isolation, which fundamentally eliminates potential risks such as electrolyte leakage and thermal runaway inherent in liquid systems. In terms of operating principles, ASSBs are fundamentally consistent with traditional LIBs (**Figure 2b**) ^[42]. During the charging stage, lithium ions are deintercalated from the lattice of the cathode and migrate through the SE to the anode, while electrons flow through the external circuit and combine with lithium ions at the anode. The discharge process reverses this sequence, with electrons outputting energy through the external circuit to power electronic devices. With the introduction of SEs, components such as liquid electrolytes, electrolyte salts, separators, and binders are no longer required, significantly

simplifying the battery manufacturing process and reducing safety risks.

2.1 Advantages of ASSBs

ASSBs, as the core direction of next-generation energy storage technology, are driving a profound transformation of electrochemical energy storage systems from liquid to solid states through the introduction of SEs. Compared to traditional liquid LIBs, ASSBs demonstrate systematic advantages in safety, energy density, cycle life, and environmental adaptability, which are considered the key technological pathway to achieving both high energy density and inherent safety [2,12,43,44].

1) High Safety Performance

The safety concerns inherent in traditional liquid electrolytes are effectively mitigated in ASSBs through the use of non-flammable, non-volatile SEs. SEs possess high thermal stability without leakage or swelling, significantly reducing risks such as electrolyte leakage, thermal runaway, and combustion [45,46]. Given the inherent non-flammability of SEs, batteries tend to resist thermal runaway when subjected to mechanical impact or thermal stimulation. This characteristic has been validated through multiple thermal safety assessments [6]. Furthermore, since SEs have high mechanical modulus and structural integrity, they can partially suppress lithium dendrite formation and penetration, which reduces the risk of internal short circuits and provides enhanced safety for LMAs. This multi-level safety design offers an ideal balance between high energy density and high reliability [47].

2) High Energy Density

ASSBs derive the energy density advantage primarily from two aspects: a wide electrochemical stability window and compatibility with LMA [12,48]. SEs have a broader stable voltage window (typically exceeding 5 V), enabling good matching with high-voltage CAMs (such as high-nickel NCM or lithium-rich manganese-based materials), reducing cathode/electrolyte interface side reactions as well as improving high-voltage operational stability [49]. Simultaneously, SEs enable the safe utilization of LMA (theoretical specific capacities as high as 3860 mAh/g), significantly boosting battery energy density. The use of LMA theoretically elevates battery energy density

far beyond liquid-based systems with liquid systems employing graphite anodes [50]. Additionally, ASSBs enhance volumetric energy density through integrated structural design. The SE simultaneously functions as a separator, enabling more compact cell design by reducing the proportion of inactive materials (such as conventional separators, electrolytes, and additional packaging). It also allows direct series connection of cells via bipolar stacking techniques, thereby simplifying battery structure and improving overall energy utilization. These design advantages not only increase energy density but also provide the structural foundation for battery modularization, miniaturization, and high-power-density applications.

3) Long Cycle Life

SEs also demonstrate significant advantages in cycle life and interface stability. Unlike liquid electrolytes, a more stable electrochemical contact layer can form at the electrode/electrolyte interface in ASSBs, thereby reducing the rate of interfacial side reactions and enabling effective control of interface degradation [46,51]. This stable solid-solid interface could benefit from buffering the volume changes in electrode materials due to lithium insertion/extraction, which significantly reduces the risk of interfacial failure and minimizes impedance increases caused by interfacial cracks. Furthermore, the high mechanical modulus of SEs helps maintain the integrity of interfacial contacts, effectively suppressing microcrack propagation during cycling. This stable interface reduces structural degradation and capacity decay during cycling to a lower level, and leads to a flatter decay curve and higher reversible coulombic efficiency under long-cycle conditions.

4) Wide Temperature Adaptability

In contrast to traditional liquid electrolytes, SEs have a much wider operating temperature range and significantly superior adaptability [52,53]. While liquid electrolytes solidify or dramatically increase in viscosity at low temperatures and volatilize or decompose at high temperatures, limiting battery applications in extreme environments [54]. SEs do not suffer from liquid phase transition issues, which can maintain reasonable ionic conductivity even below $-30\text{ }^{\circ}\text{C}$ as well as structural and

performance stability at high temperatures ($>150^{\circ}\text{C}$) without thermal runaway phenomenon. This wide temperature range capability enables ASSBs to operate reliably in extreme environments such as polar environments, aerospace applications, deep-sea exploration, and high-temperature industrial energy storage. Furthermore, since the thermal management requirements are minimal, the battery system only necessitates simple thermal control measures. This reduces overall system costs and enhances energy efficiency, providing a broader platform for diverse applications.

5) Structural Design Flexibility

Thin-film, flexible, and stacked configurations are achievable with SEs, which provide greater design flexibility to accommodate diverse application requirements [55,56]. For instance, thin-film SEs can power micro-energy systems, flexible ASSBs can be integrated into wearable devices, while stacked structures are suitable for large-scale energy storage systems [46]. Furthermore, the ASSBs eliminate the need for complex electrolyte filling and encapsulation processes, simplifying manufacturing workflows and enhancing assembly efficiency and production yield. The convenience and modularity during battery integration provide crucial technical support for the rapid development of novel energy storage systems, system lightweighting, and high energy utilization.

2.2 SE materials for ASSBs

As the core component of ASSBs, the performance of SEs directly determines the energy density, safety, cycle life, and high/low-temperature adaptability of the battery. The ideal SE should feature the following characteristics [20]: lithium-ion conductivity exceeding $10^{-3} \text{ S} \cdot \text{cm}^{-1}$ at RT, lithium-ion mobility approaching 1, wide electrochemical window ($>4.5 \text{ V}$ vs Li^{+}/Li), low electronic conductivity, outstanding chemical, electrochemical stability, good electrode interface compatibility, and high mechanical strength. Additionally, it also has environmental friendliness, simple preparation, and low cost. However, there is no single material that can simultaneously meet all of the requirements.

Currently, SEs are primarily categorized into three major types: polymer, inorganic,

and composite. **Figure 2c** compares the performance characteristics of various SEs [57]. Among these, inorganic SEs can be further subdivided into oxide, sulfide, and halide. Polymer electrolytes typically utilize PEO, PAN, or PVDF as the matrix, forming ionic conduction networks through coordination with lithium salts such as LiTFSI, LiClO₄, and LiDFO₄ [58,59]. Polymer electrolytes have the advantages of light weight, good flexibility, and easy processing into films. However, the ionic conductivity at RT is relatively low ($10^{-7} \sim 10^{-5} \text{ S} \cdot \text{cm}^{-1}$), which needs to be heated above the glass transition temperature to achieve effective ionic conduction. Meanwhile, the narrow electrochemical window and limited mechanical strength make it difficult to meet the requirements of high-energy ASSBs. Oxide SEs, represented by garnet-type, perovskite-type, and NASICON-type structures, exhibit high thermal stability, wide electrochemical windows ($>5 \text{ V}$), and excellent chemical inertness, effectively suppressing lithium dendrite growth [60]. However, the relatively low ionic conductivity ($\sim 10^{-4} \text{ S} \cdot \text{cm}^{-1}$), high preparation temperatures, and brittle nature of the materials lead to high interfacial mass transfer impedance, processing difficulties, and elevated costs [61]. Halide SEs have garnered attention for their high oxidation stability and excellent compatibility with high-voltage CAMs. However, the limited ionic conductivity and instability with LMA typically necessitate the compounding of sulfide SEs to enhance overall performance [62,63]. Sulfide SEs are considered the most promising SEs for applications given the high ionic conductivity ($10^{-4} \sim 10^{-2} \text{ S} \cdot \text{cm}^{-1}$), excellent plasticity, and low processing temperatures [64,65]. Compared to oxides, sulfides feature a larger radius and lower electronegativity of S^{2-} , which have weaker binding effects on Li^+ ions and enhance free ion concentration and migration rates. Furthermore, sulfide SEs exhibit strong compositional tunability, allowing further optimization of conductivity and stability through doping, composites, and interfacial modifications.

2.3 Superior Properties of Sulfide SEs

By comparing the RT ionic conductivities of different SE types (**Figure 2d**), it is clear that the ionic conductivity of some sulfide SEs has already surpassed that of commercial liquid electrolytes [50,66]. Liquid electrolytes and polymer SEs have lower

lithium content per unit volume but can adjust ionic conductivity across a broader lithium concentration range. In contrast, inorganic SEs require more lithium content by at least two orders of magnitude to maintain high conductivity, while the effective lithium concentration range is narrower [5]. This indicates significant differences in ionic transport mechanisms among electrolytes, closely related to the internal crystal structures. Given the technical requirements for high energy density and high safety in power lithium batteries within the new energy vehicle industry, sulfide SEs offer distinct advantages for ASSBs owing to their superiority.

1) Rapid Ion Transport

Due to the unique crystal structure and chemical composition, sulfide SEs exhibit exceptionally high ionic conductivity at RT, which typically exceeds most oxide and polymer SEs. For instance, the ionic conductivity of the LGPS SE can reach $10^{-2} \text{ S} \cdot \text{cm}^{-1}$, comparable to common liquid electrolytes [64,67]. Through doping and crystal design, ionic conductivity can be further enhanced to even higher levels [68,69]. This high conductivity facilitates reduced internal battery resistance, improved high-rate performance, and accelerated charging/discharging capabilities. The exceptional ionic conductivity of sulfide SEs primarily derives from the larger atomic radius, lower electronegativity, and weaker anion binding energy of the sulfide anion. These factors collectively reduce the energy barrier for Li^+ migration, thereby enabling highly efficient ionic transport pathways [70]. Within diverse structural frameworks such as argyrodite and glass-ceramic types, Li^+ can achieve rapid diffusion through intricate three-dimensional channel networks, resulting in significantly superior ionic conductivity compared to conventional oxide SEs. Furthermore, strategies such as doping with heteroatoms, constructing solid solutions, or designing multiphase interfaces can further regulate lattice defects and free volume in sulfide SEs, which optimize lithium-ion migration pathways and enhance ionic conductivity [71].

2) Good Workability and Plasticity

Compared to conventional oxides, sulfide SEs display significantly superior mechanical plasticity and low Young's modulus, enabling high density and excellent

interfacial contact through cold pressing at RT without high temperature sintering ^[18]. Experimental results demonstrate that sulfide SEs densify under conventional pressure conditions, which leads to good solid-solid contact with both cathode and anode. This is crucial for reducing interfacial contact resistance and improving battery performance. In contrast, oxide SEs typically require high-temperature sintering (>1000 °C) to achieve a dense state, and the high stiffness can easily lead to interfacial cracks and poor contact issues. The lower mechanical stiffness of sulfide SEs can buffer electrode volume changes, effectively reducing the risk of interfacial failure caused by mechanical stress during cycling ^[30]. Furthermore, the flexibility and compressibility of sulfide SEs enable them to effectively expand the contact between the electrolyte and active material when constructing composite electrodes with various CAMs. This enhances interfacial ion transport efficiency and makes a positive contribution to improving the energy density of large-area, high-load solid-state batteries ^[72].

3) Great Interface Compatibility

Sulfide SEs are inherently non-flammable and highly thermally stable compared to liquid electrolytes, remarkably enhancing the safety of ASSBs ^[15]. When paired with LMA, sulfide SEs provide stable lithium deposition at high critical current densities with low interfacial impedance. This helps mitigate lithium dendrite formation, thereby improving cycle stability and safety performance. The interface compatibility between sulfide SEs and CAMs has also been extensively studied ^[35]. Incorporating sulfide SEs into composite cathodes not only provides broad ion transport pathways but also offers flexible interfaces that effectively accommodate volume changes in CAMs during charging and discharging. This reduces the formation of interface cracks and helps maintain long-term interface stability during cycling. From the perspective of battery design, sulfide SEs can be directly integrated with various cathode and anode materials through cold pressing or low-temperature processing techniques. This facilitates the simplification of ASSBs manufacturing processes and enhances production scalability. Furthermore, through interface engineering optimization, such as interface modification layers and composite electrolyte structure design, side reactions between

sulfide SEs and high-voltage CAMs can be effectively mitigated, which further improves the cycling performance and energy efficiency of ASSBs.

4) Scalable Preparation

Sulfide SEs can be prepared through various low-temperature or ambient-temperature processes, including high-energy ball milling, solid-state reactions, and mechanical synthesis, which provide strong potential for scalable production^[73]. These processes not only reduce manufacturing energy consumption but also enable the fabrication of sulfide films and composite SEs, thereby meeting the demand for ultra-thin SE layers in high-energy-density batteries^[55,56]. In terms of practical application prospects, sulfide SEs are considered one of the most promising materials for realizing next-generation high-safety, high-energy-density ASSBs due to the high ionic conductivity, excellent mechanical contact properties, and interfacial compatibility. Further research on sulfide SEs is focusing on interface engineering, air stability improvement, and the development of low-cost preparation routes. These studies will provide the theoretical and technological foundation for the commercialization of sulfide SEs in ASSBs.

3 Intrinsic Stability of Sulfide SEs

Sulfide SEs are regarded as the focus material for ASSBs due to the high ionic conductivity. However, from the perspective of mass production and application, the stability of sulfide SEs in practical applications requires much attention. The stability issues of sulfide SEs can be categorized into humidity, solvent, electrochemical, thermal, and mechanical stability aspects.

3.1 Air Stability

Sulfide SEs are widely recognized as one of the key material systems for achieving high-energy-density ASSBs due to their high lithium-ion conductivity at RT and low-temperature compressible processing characteristics^[74]. However, the high sensitivity to moisture and oxygen in the air makes sulfide SEs prone to rapid chemical degradation under environmental exposure, which becomes one of the core obstacles limiting the large-scale preparation and engineering applications^[24]. Extensive research

indicates that when sulfide SEs contact water molecules, significant hydrolysis and accompanying oxidation reactions occur [75-77]. This leads to irreversible destruction of the crystal lattice structure and releases corrosive and toxic gases such as H₂S. Such degradation typically involves obstruction of lithium-ion transport pathways, causing a sharp decline in ionic conductivity and a significant increase in interfacial impedance. Consequently, the electrochemical performance of the battery is severely degraded. From the microstructural perspective, the air instability of sulfide SEs primarily stems from the chemical bond characteristics. The P-S bond or M-S bond (where M represents electroactive or structurally stable cations such as Ge, Sn, or Si) within the material exhibits lower bond energy and stronger polarizability compared to the M-O bond in oxide, making it susceptible to attack by water molecules or oxygen [78]. The sulfur anion (S²⁻) in the lattice exhibits typical soft-base behavior, which has a strong affinity for protons or polar molecules in the environment. This significantly lowers the kinetic barrier for hydrolysis reactions and accelerates the decomposition of the sulfide structure. Consequently, the powder synthesis, electrode composite preparation, and cell assembly processes for sulfide SEs in current research and engineering practice typically require strict control of moisture and oxygen content under inert atmospheres. This high dependence on environmental conditions not only substantially increases manufacturing costs but also poses a substantial barrier to the industrialization of sulfide-based ASSBs [20-23].

Based on the hard and soft acids and bases (HSAB) theory, researchers have systematically elucidated the fundamental mechanism underlying the air instability of sulfide SEs [78]. The S²⁻ acting as a typical soft base, readily interacts with hydrogen (H), which exhibits soft acid characteristics in water molecules. This interaction triggers the breaking of P-S or M-S bonds, leading to the generation of volatile hydrogen sulfide. Simultaneously, oxygen undergoes oxidation reactions with lithium within the lattice, which induces the formation of byproducts such as Li₂O and LiOH. This process triggers local structural rearrangements or interfacial phase transitions, leading to a continuous deterioration of interfacial transport properties during electrochemical

cycling ^[79]. From a structural perspective, the low lattice energy, insufficient charge density, and highly polar covalent bond characteristics collectively constitute the root causes of the thermodynamic and kinetic susceptibility of sulfide SEs to hydrolysis and oxidation reactions. Based on the HSAB theory, further developed theoretical models provide tools for quantitatively understanding the air degradation behavior of sulfide SEs. The Hard-Soft-Electron-Hole theory proposed by Mulks et al. extends the HSAB concept to multi-electron systems ^[80]. By introducing electron and hole behavior analysis, the theory offers a more universal perspective for explaining the electronic structure evolution and chemical destabilization of sulfide lattices under environmental stimuli.

For enhancing the air stability of sulfide SEs, researchers have proposed multiple material design strategies from both lattice chemistry regulation and surface engineering perspectives in recent years. The first strategy enhances lattice bond energy through elemental doping or ion substitution to reduce the reactivity of sulfur anions toward water molecules. For instance, the partial introduction of O²⁻ into sulfide structures forms the more thermodynamically stable P-O bond, thereby mitigating the chemical instability of the original P-S bond and significantly reducing the hydrolysis rate (**Figure 3a**) ^[81-83]. Alternatively, replacing original lattice sites with cations possessing stronger metal-sulfur bonding energy increases overall lattice hardness and reduces bond polarizability, which enhances the environmental stability of the material (**Figure 3b**) ^[84]. The second strategy focuses on surface engineering design, physically blocking the ingress of moisture and oxygen by constructing dense inorganic protective layers or hydrophobic coatings on the surfaces of sulfide particles. Research indicates that coatings of oxides, or the introduction of hydrophobic organic-inorganic hybrid layers, can effectively suppress H₂S release and significantly delay the structural degradation of sulfide SEs, thereby improving the stability under air exposure (**Figure 3c-d**) ^[36,85-87]. The third approach improves the environmental compatibility of sulfides through overall composition optimization and defect control. By adjusting the cation-anion ratio, introducing controlled defects, or optimizing the local structural

environment, the adsorption capacity of water molecules on the lattice surface can be reduced. Alternatively, the material can be prompted to form a relatively stable passivation layer upon initial air contact, thereby achieving a synergistic preservation of structural integrity and ionic conductivity ^[17,72].

Despite positive progress in both experimental and theoretical studies, the inherent air sensitivity of sulfide SEs remains fundamentally unresolved. Future material design must prioritize systematic regulation of HSAB chemistry while developing theoretical models capable of quantifying air stability predictions. These models should comprehensively describe bond-energy strengthening mechanisms, multiscale interfacial phase-transition kinetics, electron-hole coupling behavior, and solid-gas interface reaction thermodynamics. Simultaneously, balancing enhanced air stability with high ionic conductivity, favorable electrode interface compatibility, and acceptable mass production costs remains a core scientific and technological challenge for sulfide-based ASSBs advancing toward engineering applications. Overall, deepening the fundamental understanding of the air degradation mechanisms in sulfide SEs and proposing verifiable, predictable stabilization strategies will lay a solid theoretical foundation and engineering pathway for the industrialization of sulfide-based ASSBs.

3.2 Solvent Compatibility

In the wet-process manufacturing stages (such as electrode slurry preparation, SE membrane processing, and battery assembly), solvent compatibility refers to the ability of SEs to maintain the crystalline integrity and electrochemical stability of the electrolyte when interacting with solvent media (e.g., N-methyl pyrrolidone, water, and other organic solvents). Given the high maturity of wet-process coating and slurry processing techniques within the existing LIB industry chain, solvent compatibility has emerged as a key constraint limiting the large-scale manufacturing and engineering applications of sulfide SEs ^[88]. Solvent intolerance not only triggers material structural degradation and interfacial failure but also significantly inflates production costs and reduces battery consistency, thereby restricting the compatibility of sulfide-based ASSBs with existing manufacturing systems ^[24]. Extensive research indicates that

typical sulfide SEs (such as Li_3PS_4 , $\text{Li}_7\text{P}_3\text{S}_{11}$, LGPS, LPSCl, etc.) exhibit significant chemical sensitivity to polar solvents [89,90]. In polar organic solvents, even trace amounts of water can induce rapid hydrolysis of sulfides, generating H_2S gas and insulating byproducts like Li_2O and LiOH . This leads to a sharp decline in ionic conductivity and disrupts the continuity of ion transport between particles. Furthermore, N-methyl pyrrolidone, as a widely used industrial solvent, has a high dielectric constant and strong solvation ability. It irreversibly reacts with PS_4^{3-} or $\text{P}_2\text{S}_7^{4-}$ structural units in sulfides, causing P-S bond breakage and structural unit rearrangement. This generates decomposition phases with low ionic conductivity, severely degrading both bulk and interfacial transport properties of the electrolyte [91]. The degradation behavior of sulfide SEs in different solvents is closely related to solvent polarity and molecular structure (**Figure 4a**) [92]. Polar solvents with high dielectric constants and electronegative atoms such as N or O typically exhibit stronger chemical erosion capabilities, whereas nonpolar or weakly polar solvents (e.g., toluene, heptane) significantly reduce the destructive effect on sulfide lattices [93]. Based on the HSAB theory, the central atom P^{5+} in sulfide SEs belongs to relatively soft Lewis acids, while polar solvent molecules containing lone pairs of electrons can act as strong Lewis bases and attack nucleophilically, inducing P-S bond cleavage and generating low-conductivity structural units (**Figure 4b**) [24]. In contrast, solvents possessing stable aromatic ring structures and lacking strong coordinating functional groups exhibit greater chemical inertness, thereby demonstrating superior compatibility with sulfide SEs [94].

To mitigate solvent instability in sulfide SEs during wet processing, researchers have proposed multiple control strategies at both the process and material levels. At the process level, selecting low-polarity or non-polar solvents as slurry dispersion media can partially suppress solvent-induced chemical decomposition [95]. Concurrently, the development of dry electrode fabrication techniques offers a novel technical pathway to entirely circumvent solvent-related issues [96]. At the material level, enhancing the intrinsic stability of sulfide SEs through composition design is considered a more

universal solution. For instance, incorporating stabilizing components like InF_3 enhances lattice bond energy and reduces polarizability, which enables modified sulfide SEs to maintain high ionic conductivity even after organic solvent immersion (**Figure 4c**)^[97]. Furthermore, constructing solvent-compatible binder systems and employing cosolvent strategies have also been demonstrated to help preserve the structural and performance stability of sulfide SEs under wet processing conditions^[98]. In general, the solvent compatibility issue of sulfide SEs originates from the inherent mismatch between the lattice chemistry and polar solvents, representing one of the core bottlenecks limiting the compatibility with existing LIB manufacturing processes. Future research must advance in areas such as solvent screening, intrinsic stability design of materials, and synergistic optimization of manufacturing processes. This will enable the reliable application of sulfide SEs in large-scale manufacturing, which lays the foundation for the engineering and industrialization of sulfide-based ASSBs.

3.3 Thermal Stability

Thermal stability of sulfide SEs should be systematically evaluated from two perspectives: the thermal decomposition behavior of the electrolyte itself and the interfacial thermal reactions between the electrolyte and electrode^[99,100]. From the intrinsic material perspective, most crystalline sulfide SEs exhibit a high thermal stability maximum. Under inert atmospheres, representative materials such as LGPS, LPSCl, and $\text{Li}_7\text{P}_3\text{S}_{11}$ maintain stable crystal structures within the 400-500 °C range without significant thermal decomposition or phase transitions, which demonstrates thermal resistance significantly superior to conventional carbonate-based liquid electrolytes (**Figure 5a**)^[99,101,102]. This elevated thermal decomposition temperature primarily results from the continuous covalent-ionic hybrid bond network within the sulfide lattice and the low content of volatile components, providing enhanced structural retention under thermal excitation. However, the thermal safety of sulfide SEs in practical operation or thermal runaway scenarios of ASSBs is often constrained by the interfacial thermal stability with electrode materials, particularly high-energy-density CAMs^[103]. Research indicates that when battery temperatures rise to

1 approximately 200-300 °C, high-nickel layered oxide CAMs (such as
2 LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂) are prone to lattice instability during charging, which releases
3 reactive oxygen species and strongly exothermic reactions with sulfide SEs [28,104]. This
4 process typically involves P-S bond cleavage and oxidation of sulfur species, yielding
5 multiple reaction products including P₂S_x, SO_x, and metal sulfides. Moreover, it
6 significantly amplifies the total heat release of the system, thereby reducing the thermal
7 safety margin of ASSBs [105].

8
9 The microstructure at the interface plays a crucial regulatory role in the occurrence
10 pathways and reaction intensity of thermal reactions. Research indicates that applying
11 higher external pressure during the forming process of composite cathodes facilitates
12 the in-situ formation of a dense amorphous P₂S_x reaction layer at the cathode/sulfide
13 interface (**Figure 5b-c**) [99,106,107]. This layer can partially block the diffusion of reactive
14 oxygen species toward the electrolyte side and suppress further interfacial
15 decomposition reactions, thereby reducing the overall heat release by approximately
16 40-50% (**Figure 5d-f**) [108,109]. Such a phenomenon shows that moderately controlled
17 interfacial reactions are not entirely detrimental but may enhance thermal stability by
18 constructing kinetically stable interface layers. Beyond structural regulation, material
19 composition design also effectively improves sulfide thermal stability. Introducing
20 oxygen into sulfide lattices to form oxysulfide SEs increases P-O bond ratios and
21 enhances overall bond energy, thereby reducing decomposition tendencies at elevated
22 temperatures [89,110]. Concurrently, introducing thermally stable oxide coatings (e.g.,
23 LiNbO₃ or Al₂O₃) onto cathode particle surfaces effectively buffers oxygen evolution
24 and inhibits direct contact with sulfide SEs, significantly delaying the onset of
25 interfacial exothermic reactions [85,111,112]. Therefore, sulfide SEs exhibit excellent
26 internal thermal stability. However, the thermal reactions at the interface with high-
27 voltage CAMs remain a critical factor limiting the thermal safety of ASSBs. Future
28 research should advance synergistically across multiple dimensions, which include
29 electrolyte composition design, interface structure regulation, and overall battery
30 thermal management strategies. This approach aims to systematically improve the
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1 thermal safety margins of sulfide-based ASSBs while maintaining high ionic
2 conductivity and interface stability.
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4 *3.4 Electrochemical Stability*

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6 The electrochemical stability of sulfide SEs is primarily constrained by the narrow
7 intrinsic thermodynamic stability window, which remains one of the core challenges
8 limiting the application of these materials in high-energy-density ASSBs (**Figure 6a-b**)
9 [21,74,113]. First-principles calculations reveal that the true thermodynamic stability range
10 of typical sulfide SEs is significantly narrower than the apparent electrochemical
11 window obtained experimentally via methods such as cyclic voltammetry (**Figure 6c**)
12 [114]. For instance, the intrinsic stable potential range of LGPS is only approximately
13 1.7-2.1 V (vs. Li^+/Li), markedly lower than the operating voltage range it endures in
14 actual batteries. Below the lower limit, sulfide SEs undergo reductive decomposition to
15 form thermodynamically stable products like Li_2S and Li_3P [33,115]. Above the upper
16 limit, oxidative reactions occur to generate sulfide or polysulfide phases such as P_2S_5
17 and GeS_2 . Both reactions possess significant thermodynamic driving forces, indicating
18 that sulfide SEs are inherently metastable across a wide voltage range. Nevertheless,
19 sulfide SEs in practical ASSBs often operate under conditions far exceeding the
20 thermodynamic stability window, primarily attributed to the kinetic stabilization effect
21 induced by decomposition products at the interface. The interfacial film formed after
22 initial decomposition of the electrolyte significantly alters subsequent reaction
23 pathways and rates, thereby governing the electrochemical behavior. Based on the ion
24 and electron transport characteristics of the interfacial phase, sulfide/electrode
25 interfaces can generally be categorized into three types [116]: thermodynamically stable
26 interfaces, mixed ion-electron conductive interfaces, and passivated interfaces
27 resembling SEI. Among these, passivated interfaces exhibiting both high ionic
28 conductivity and low electronic conductivity are most desirable, which can effectively
29 block continuous electrochemical decomposition reactions while maintaining ionic
30 transport continuity at the interface.
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33 On the cathode side, the operational voltage of high-potential layered oxide CAMs
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far exceeds the thermodynamic stability threshold of sulfide SEs, making interfacial instability particularly pronounced ^[117,118]. To suppress electrolyte oxidation decomposition and prevent the formation of electronically conductive byproducts, stable oxide buffer layers are typically introduced onto the cathode particle surface. Such coating layers establish a chemical potential gradient between the electrolyte and the high-potential cathode, reducing the driving force for interfacial reactions while providing stable ion transport pathways, thereby significantly enhancing cycling stability. On the anode side, the interface stability between sulfide SEs and LMA also faces severe challenges (**Figure 6d-e**) ^[33]. If decomposition products at the interface exhibit electronic conductivity, they can induce persistent side reactions and accelerate the nucleation and growth of lithium dendrites and eventually lead to internal short circuits. To avoid these issues, researchers constructed artificial interface layers to guide the formation of an ion-conductive, electron-barrier interface structure, thereby achieving effective passivation of interfacial reactions (**Figure 6f**) ^[22]. Overall, the electrochemical stability of sulfide SEs is not solely determined by the intrinsic thermodynamic properties of the electrolyte itself, but is dominated by the kinetic stability regulated by interfacial reactions. Through sophisticated interfacial engineering, constructing stable, controllable, and electronically insulating interface layers on both the cathode and anode sides represents a key scientific pathway to overcoming the bottleneck of electrochemical stability in sulfide SEs and achieving long-life ASSBs.

3.5 Mechanical Stability

Mechanical stability issues in sulfide SEs primarily stem from the chemical-mechanical coupled failure behavior during electrochemical reactions, which is considered one of the critical bottlenecks limiting the long-term reliable operation of high-energy-density ASSBs ^[119]. Compared to garnet-type oxide SEs, sulfide SEs typically exhibit lower Young's Modulus and a degree of plastic deformation capability. This characteristic facilitates close contact with electrodes under applied stacking pressure and reduces initial interfacial impedance. However, the limited fracture

1 toughness and volume effects induced by interfacial reactions make sulfide SEs more
2 susceptible to mechanical degradation during cycling. When sulfide SEs directly
3 contact LMA, the decomposition reaction inevitably occurs, which generates Li_2S , Li_3P ,
4 and related composite phases ^[120]. These decomposition products typically possess
5 molar volumes significantly different from the base phase, inducing localized volume
6 expansion at the interface region. Experimental characterization and mechanical
7 modeling results indicate that such non-uniform volume changes generate significant
8 tensile stress concentrations within the SE particles, serving as the primary driving force
9 for microcrack nucleation ^[108,121]. The elastic strain energy accumulating within larger
10 sulfide particles (typically exceeding $\sim 3\text{ }\mu\text{m}$) increases substantially. Once this exceeds
11 the fracture toughness threshold of the material, cracks propagate rapidly and lead to
12 particle fragmentation ^[74]. Particle-level fracture not only disrupts the continuous ionic
13 conduction network but also creates low-impedance pathways within the electrolyte,
14 which provide routes for lithium dendrite penetration and growth, ultimately inducing
15 internal short-circuit failure ^[22,32].

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17 By reducing the particle size of sulfide SEs, chemical-mechanical coupling failure
18 can be effectively mitigated. Research indicates that when particle size is reduced to
19 the submicron scale (typically less than $1\text{ }\mu\text{m}$), the volume expansion caused by
20 interfacial reactions is more readily distributed uniformly within the particles, thereby
21 significantly reducing stress concentration levels ^[72]. Under these conditions, sulfide
22 particles exhibit a greater tendency toward coordinated deformation rather than
23 catastrophic crack propagation even when some interfacial decomposition occurs,
24 thereby maintaining structural integrity and extending cycle life. From the perspective
25 of intrinsic mechanical properties, the Young's Modulus of sulfide SEs typically ranges
26 between 10-30 GPa, significantly lower than that of typical oxide SEs ^[66]. The lower
27 modulus can help to buffer interfacial contact stresses and improve electrode-
28 electrolyte adhesion ^[122]. However, sulfide SEs generally exhibit low fracture toughness
29 (typically only $0.2\text{-}0.4\text{ MPa}\cdot\text{m}^{1/2}$), which makes them more susceptible to brittle
30 fracture when preparing ultrathin SE films or enduring cyclic stress fluctuations

(**Figure 7a**)^[123,124]. Therefore, relying solely on the intrinsic mechanical properties of the material makes it difficult to simultaneously meet the demands for high interfacial contact and high crack resistance. To systematically enhance the mechanical stability of sulfide SEs, researchers have proposed various optimization strategies at both the structural and material levels. Among these, constructing sulfide-polymer composite SEs is considered a proven effective approach. By introducing flexible polymer components into the sulfide continuous phase, localized stress concentrations induced by interfacial reactions can be effectively dispersed, which significantly enhances overall fracture toughness while maintaining acceptable ionic conductivity under proper design (**Figure 7b**)^[125-127]. Furthermore, optimizing battery assembly pressure ensures dense interfacial contact while preventing stress accumulation from excessive mechanical constraints. From a materials chemistry perspective, developing novel sulfide compositions with reduced volumetric changes during interfacial reactions or regulating decomposition pathways through doping are also recognized as key approaches to mitigate chemical-mechanical coupling failure risks (**Figure 7c-d**)^[56,128,129]. In summary, the mechanical stability of sulfide SEs is not determined by a single material parameter but results from the combined effects of interfacial chemical reactions, microstructural characteristics, and external mechanical constraints. Through particle size regulation, composite material design, and synergistic optimization of battery structure and assembly conditions, it is possible to fundamentally mitigate mechanical failure issues in sulfide SEs, providing critical support for their application in high-performance ASSBs.

4 Interface of Sulfide/Cathode

4.1 Interface Issues

The stability of the interface between sulfide SEs and CAMs is widely recognized as one of the critical bottlenecks that limit high energy density and long-term operation in batteries. The interface on the cathode side must simultaneously withstand high operating voltages, strongly oxidizing conditions, and significant chemical potential gradients, making its evolution process more complex. Extensive research indicates that

sulfide SEs exhibit inherent defects at multiple levels, such as chemical stability, electrochemical stability, and mechanical contact stability when in direct contact with high-voltage oxide CAMs. This leads to rapid increases in interfacial impedance and sustained degradation of battery performance.

4.1.1 Interfacial Chemical/Electrochemical Instability

Significant chemical and electrochemical instability at the interface of sulfide SEs and CAMs is the primary issue that limits high-voltage operation in batteries. Sulfide SEs typically possess a narrow electrochemical stability window, with oxidation decomposition potentials often below 5 V (vs. Li/Li⁺), which is significantly lower than the actual operating potentials of layered oxide CAMs at high states of charge^[49]. When sulfide SEs directly contact CAMs and operate under high-voltage conditions, oxidative decomposition reactions inevitably occur at the interface. This decomposition process typically initiates with the preferential oxidation of S²⁻ anions, along with the destruction of the P-S framework structure and the generation of byproducts such as elemental sulfur, polysulfides, phosphor sulfides, and transition metal sulfides^[130,131]. These products are predominantly electronic or ionic insulators and accumulate progressively at the interface, significantly impeding lithium-ion migration across the interface and causing rapid increases in interfacial polarization and impedance. Both first-principles calculations and experimental studies indicate that such reactions possess a pronounced thermodynamic driving force, which makes them difficult to completely suppress through simple process optimization (**Figure 8a**)^[99,132]. The oxidation potential of sulfide SE is only approximately 2 V, making it highly susceptible to decomposition reactions in high-voltage cathode environments. The mixed-phase layers formed at the interface (such as Li₂S, Li₄GeS₄, and P₂S_x) exhibit limited ionic conductivity and high electronic conductivity, which demonstrates mixed ion-electron conductor characteristics (**Figure 8b**)^[116,133,134]. Such interfacial layers not only directly elevate the initial interfacial impedance but also provide pathways for electron transport across the interface. This continuously induces further oxidative decomposition of the electrolyte, leading to cumulative and irreversible degradation at

the interface.

4.1.2 Interfacial Reactions Under High Voltage

Under high-voltage conditions, the chemical reactivity of CAMs significantly increases and causes synergistic damage to the stability of the sulfide SE interface ^[135]. Particularly in high-nickel layered oxide CAMs, the lattice oxygen stability markedly decreases as lithium depletion progresses, readily releasing highly reactive oxygen species (O^{2-} or O_2) with strong oxidizing properties. These reactive oxygen species can directly react with S^{2-} in the sulfide SE to form SO_x species, which accelerates the structural degradation of the electrolyte. The involvement of reactive oxygen species transforms the interface reaction from a unidirectional oxidative decomposition of the electrolyte into a bidirectional coupled reaction process between the cathode and electrolyte. On the one hand, reactive oxygen species promote non-uniform oxidation of the sulfide SE, causing the interfacial reaction layer to gradually thicken and present a complex multiphase structure. On the other hand, the transition metal oxide layer enriched with oxygen vacancies gradually forms on the cathode surface, which further reacts with the sulfide SE to generate high-impedance products such as metal sulfides and phosphorus oxides (**Figure 8c**) ^[136]. For example, significant interfacial decomposition phenomena can be observed in the LPSCI-NCM cathode system within the operating voltage range of 2.5-4.1 V (**Figure 8d**) ^[137]. Although the interface remains relatively stable initially, the continuous oxidative decomposition of the electrolyte coupled with sulfidation reactions on the cathode surface during cycling leads to the accumulation of interfacial impedance ^[5]. This phenomenon is particularly pronounced in nickel-rich NCM and NCA cathodes, which greatly limits the cycling life of high-voltage sulfide ASSBs.

4.1.3 Interface Instability Mechanism

From the electronic structure perspective, the electrochemical instability at the sulfide SE/cathode interface fundamentally originates from unreasonable energy level matching ^[22]. The electrochemical potential of the cathode decreases as the operating voltage increases. When it falls below the highest occupied molecular orbital (HOMO)

energy level of the sulfide SE, electrons transfer from the electrolyte to the cathode, which triggers the oxidative decomposition of the electrolyte. Since sulfide SEs generally possess higher HOMO energy levels, the electronic instability is more likely to occur in high-voltage CAMs ^[114]. Beyond electronic energy level factors, the significant lithium chemical potential difference between the cathode and sulfide SE also exists ^[138]. Oxide cathodes typically exhibit higher lithium chemical potentials, while the binding energy of lithium ions in sulfide SE is relatively low, causing lithium ions to spontaneously migrate from the electrolyte toward the cathode. This process induces lithium concentration redistribution at the interface and forms a lithium-depleted space charge layer, thereby reducing the effective ionic conductivity and increasing the interfacial impedance in the interface region. It should be noted that the space charge layer is not the sole source of high impedance at the interface. Chemical composition changes and byproduct accumulation at the interface often exert a more direct inhibitory effect on ion transport ^[116,133,136]. Therefore, interface instability typically results from the combined effects of energy level mismatch, chemical potential-driven lithium migration, and the growth of chemical reaction layers.

4.1.4 Chemical-Mechanical Coupled Failure

At the mechanical level, the rigid solid-solid contact between the SE and cathode further amplifies interfacial issues ^[139]. Compared to liquid electrolyte systems, solid-state systems exhibit poorer interfacial wettability and limited effective contact area, resulting in higher interfacial ion migration barriers and inherently greater interfacial impedance. This problem intensifies during cycling as chemical reactions and volume changes occur concurrently. The CAM typically undergoes structural phase transitions and volume expansion/contraction during lithium insertion and extraction, while the sulfide SE often experiences volume contraction during oxidative decomposition. This mismatched volumetric change induces stress concentration at the interface, which accumulates continuously during cycling ^[140]. Research indicates that repeated stress loading induces crack propagation within cathode particles, secondary particle fragmentation, and gradual loss of electrolyte-cathode contact, ultimately disrupting ion

transport pathways. In situ mechanical testing reveals that volume changes induced by electrolyte oxidation constitute a significant source of stress accumulation at the interface ^[141]. The coupling of chemical reactions with mechanical instability renders interface degradation irreversible, which becomes one of the key factors limiting the long-term stable operation of sulfide-based ASSBs.

4.2 Strategies for Modifying the Cathode Interface

The cathode/electrolyte interface is considered one of the core bottlenecks limiting high-voltage operation, high areal capacity output, and long cycle life. Unlike the spontaneously formed SEI in liquid systems, sulfide SE often undergoes vigorous chemical and electrochemical side reactions with high-voltage CAMs, forming high-impedance interfacial layers with severe mechanical contact degradation. To address the inherent chemical mismatch between sulfide SE and high-voltage CAM, recent research has converged on three primary solution pathways: cathode composition optimization, material coating, and novel cathode material development.

4.2.1 Optimization of Cathode Components

From the perspective of cathode composition optimization, precisely controlling the ratio as well as the structure of the sulfide SE, CAM, and conductive component within the composite cathode represents one of the core strategies for mitigating interfacial instability and enhancing electrochemical performance. The mass fraction of sulfide SE in composite cathode systems is a key parameter to determine the contact quality at the cathode/electrolyte interface and ion transport efficiency. For nickel-rich layered oxide/sulfide composite cathodes, appropriately increasing the content of sulfide SE can significantly improve the contact state at the three-phase interface ^[142]. On one hand, a continuous SE network constructs three-dimensional lithium-ion permeation pathways within the composite electrode, which reduces the tortuosity of ion migration paths. On the other hand, the flexible sulfide SE forms effective mechanical interlocking with CAM particles during compaction, thereby enhancing interfacial contact stability. This structural optimization strategy simultaneously reduces interfacial contact impedance and bulk ion migration barriers by expanding the

effective lithium-ion transport interface area. Consequently, high-nickel CAMs maintain efficient charge transport within low-porosity composite structures, providing crucial support for achieving high-rate and long-life ASSBs. However, the mass ratio between SE and CAM is not as high as better, but rather requires achieving a dynamic equilibrium between ion and electron transport (**Figure 9a**) ^[143]. When the CAM content is excessively high, the SE exhibits insufficient connectivity within the composite cathode. This fragmentation of the ion transport network leads to rapid polarization increases and capacity decay under high-rate or high-loading conditions. Conversely, when the SE proportion is excessively high, the CAM becomes overly dispersed, and the electron transport pathways become discontinuous, resulting in insufficient electron conductivity that similarly limits cathode reaction kinetics. Therefore, rationally optimizing the mass fraction of SE facilitates the simultaneous construction of continuous ion transport channels and a dense electron permeation network. This synergistically reduces interfacial charge transfer impedance, enhancing the capacity utilization and cycling stability of the cathode.

In the pursuit of high energy density, the areal capacity of ASSBs is widely recognized as needing to exceed 3 mAh cm⁻² to possess the potential to compete with liquid systems. However, the current areal capacity of most sulfide-based ASSBs remains at the 1-2 mAh cm⁻² level. The fundamental reason lies in the restricted ion and electron transport within high-loading cathodes and the exacerbated interfacial polarization ^[144]. Introducing highly ionically conductive electrolytes on the cathode side while reducing their loading represents a key strategy to achieve both high loading and high-rate performance. For instance, employing halide SEs with low interfacial impedance on the cathode side can maintain a continuous ionic transport network even at low loading levels, thereby significantly reducing interfacial polarization (**Figure 9b**) ^[145]. This strategy demonstrates that cathode optimization extends beyond increasing electrolyte content to enhancing the interface utilization efficiency per unit mass of electrolyte.

Beyond ion transport, the electronic conductivity of composite cathodes is equally

critical under high state-of-charge conditions. Most layered oxide CAMs exhibit inherently low electronic conductivity, especially pronounced in solid-state systems, thus requiring the introduction of conductive additives to establish three-dimensional electron transport networks. However, the incorporation of conductive components simultaneously exerts profound effects on cathode/sulfide interface stability ^[146]. Carbon-based conductive agents can significantly enhance electron transport efficiency, but the high electronic conductivity and large specific surface area of carbon may induce oxidative decomposition of the sulfide SE at high potentials. This generates high-impedance interfacial byproducts, thereby deteriorating interface stability. Furthermore, the type, morphology, and spatial distribution of conductive agents exert a decisive influence on interfacial behavior (**Figure 9c**) ^[147]. Traditional spherical carbon black with three-dimensional packing characteristics often displaces the effective volume fraction of the electrolyte, which narrows the lithium-ion transport channels. This locally creates regions of high electron concentration, accelerating the oxidative decomposition of the sulfide SE. In contrast, fibrous or flake-like conductive materials can reduce the spatial crowding effect through a two-dimensional arrangement. This maintains electronic conductivity while preserving a continuous ionic transport network, thereby mitigating interfacial side reactions. Moreover, the content of conductive agents requires precise regulation. Under low content conditions, the proportion of direct contact between conductive agents and sulfide SEs is relatively high, leading to more pronounced interfacial side reactions. whereas at higher contents, the continuous electronic network can significantly enhance the overall conductivity of the cathode, enabling high-loading electrodes to achieve high areal capacity even at high rates ^[148]. Gradient composition design has gradually emerged as an effective strategy for mitigating cathode interface issues (**Figure 9d**) ^[149]. By introducing spatial gradients of conductive agents or electrolyte content within composite cathodes, conditions for electron and ion transport can be optimized in distinct regions. For instance, increasing the conductive agent proportion near the current collector ensures rapid electron transport, while reducing the conductive agent content and increasing the